

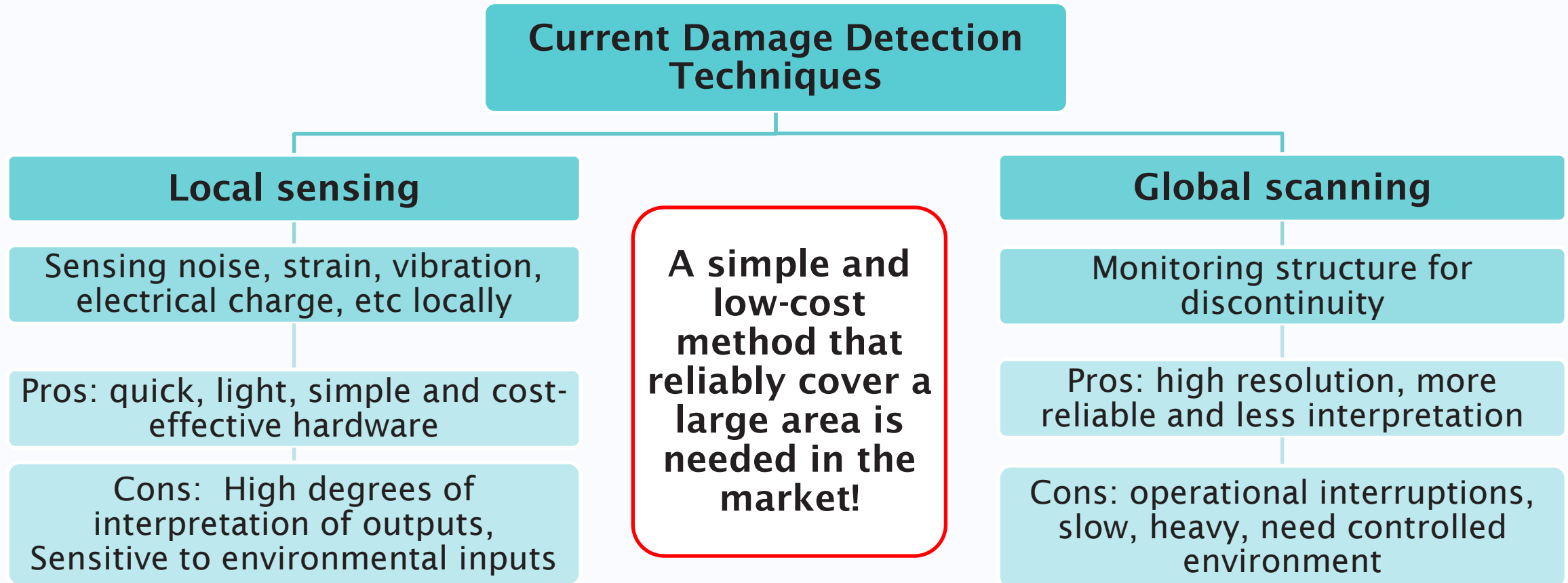
Overstrain sensors for Structural Health Monitoring

Meisam Jalalvand
Associate Professor in Composite Materials

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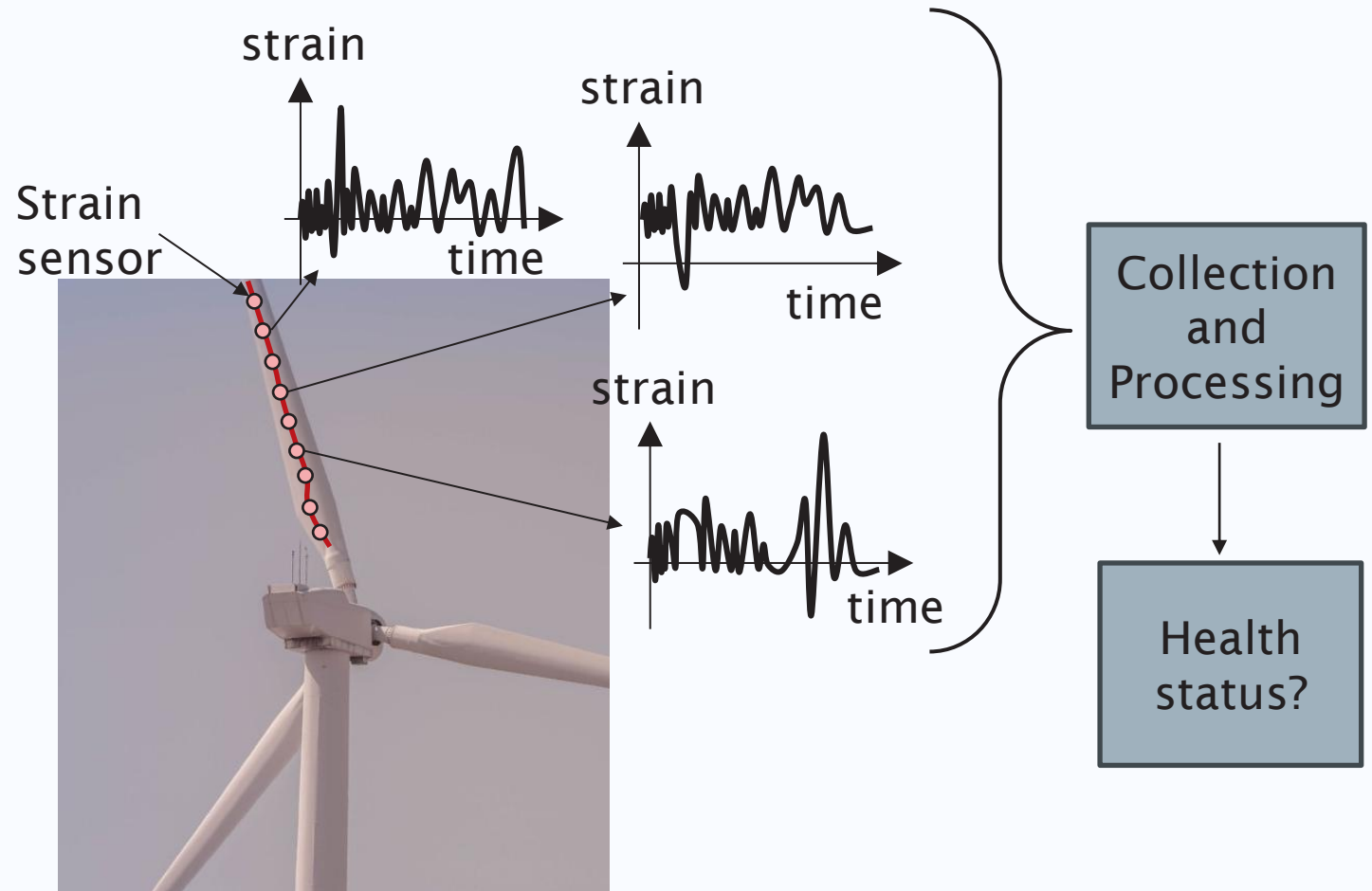


SHM State-of-the-art: complex and/or costly

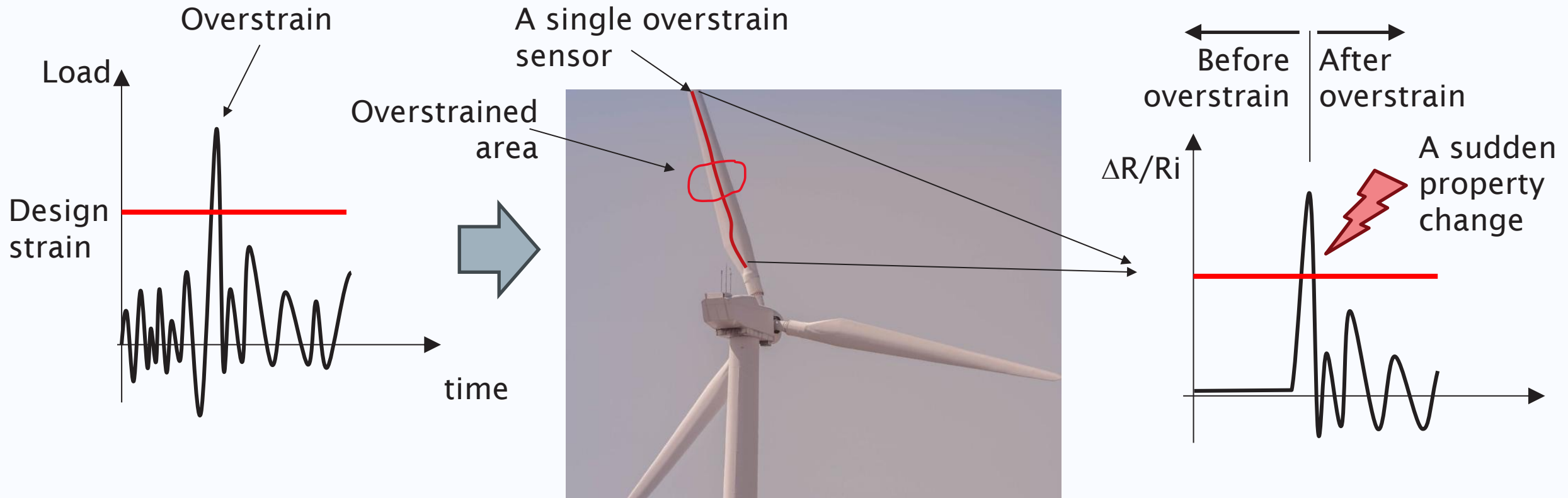


Interpretation of strain is challenging!

- Strain changes with time and location
- Needs continuous sensing
- Sensitive to environment e.g. temperature, moisture, etc,
- Generating a huge volume of data
- Often requires recalibration
- Interpretation of a structure's health from strain is challenging



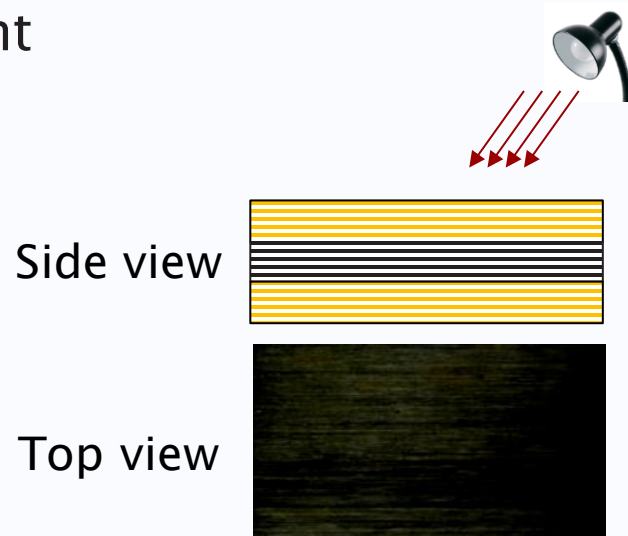
Overstrain Sensing + localising the overstrained area



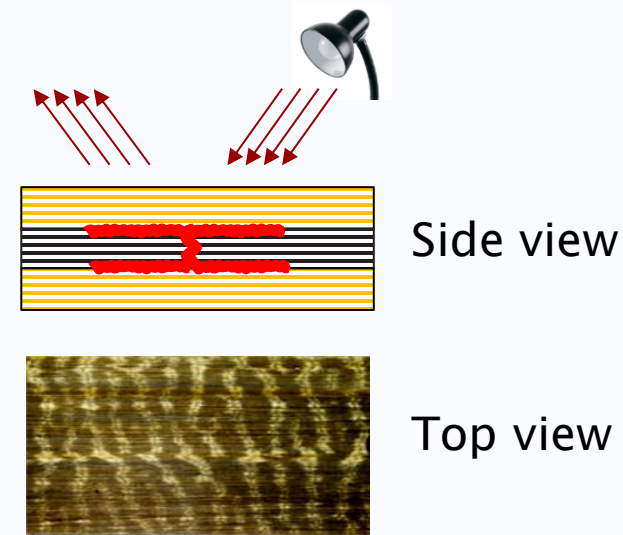
Visual Overstrain Sensor

Visual static overstrain sensor

- Carbon layer fracture and delamination produces light areas, changing the appearance of glass/carbon hybrids.
- UK Patent



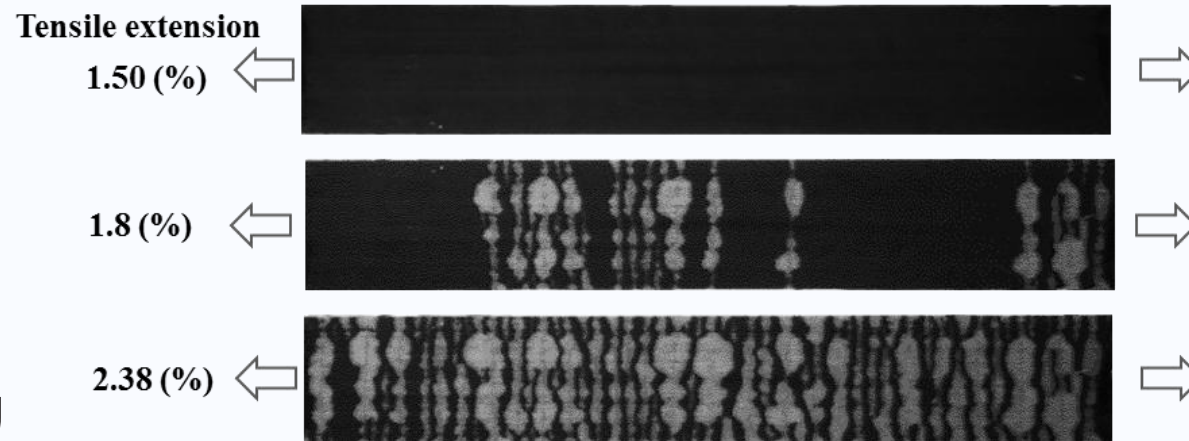
Light is completely absorbed by the carbon layer.



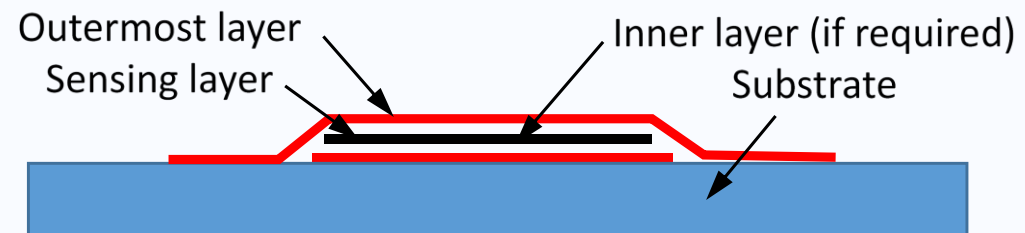
Delamination and gap at glass/carbon interface reflects light back.

Visually detectable damage

- Wireless and offline
- Low cost
- Robust
- Lightweight
- Requires no special training
- Whole surface layer or a patch
- Directional



Ordinary photos, NOT special NDT



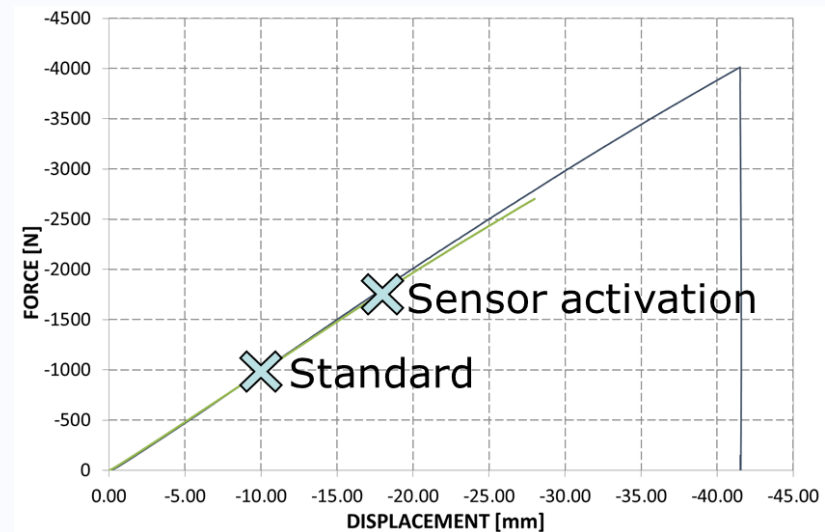
Handlebar instrumented by the overstrain sensor



EN14781 Standard load:
1000 N

Applied load: 2700 N

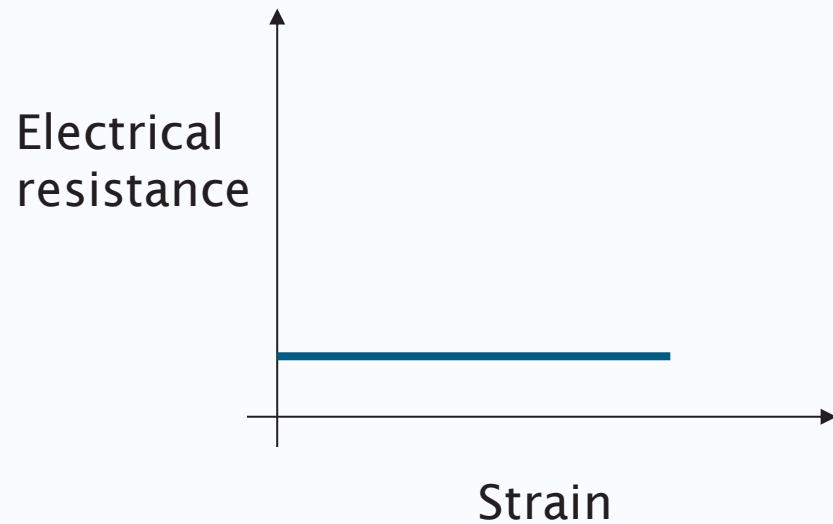
Damage visualised at:
1750 N



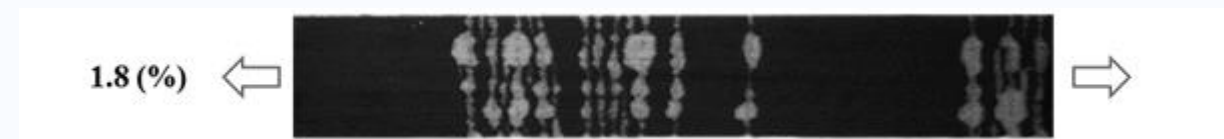
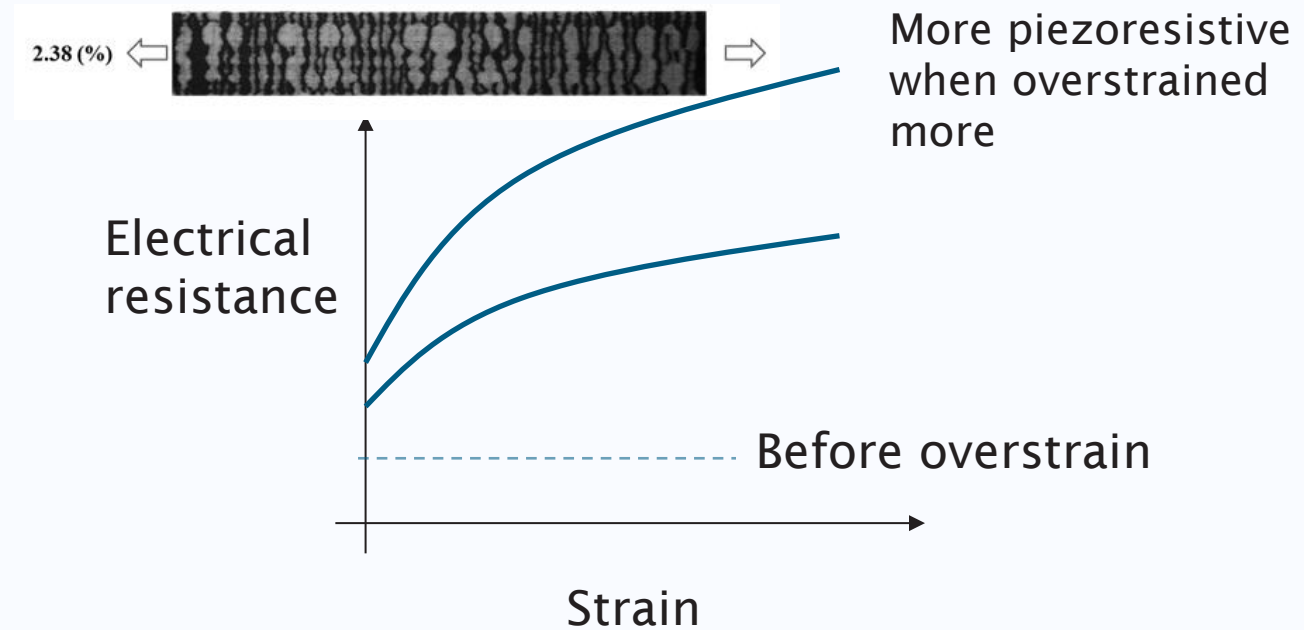
Remote Overstrain Sensor

Sensing mechanism

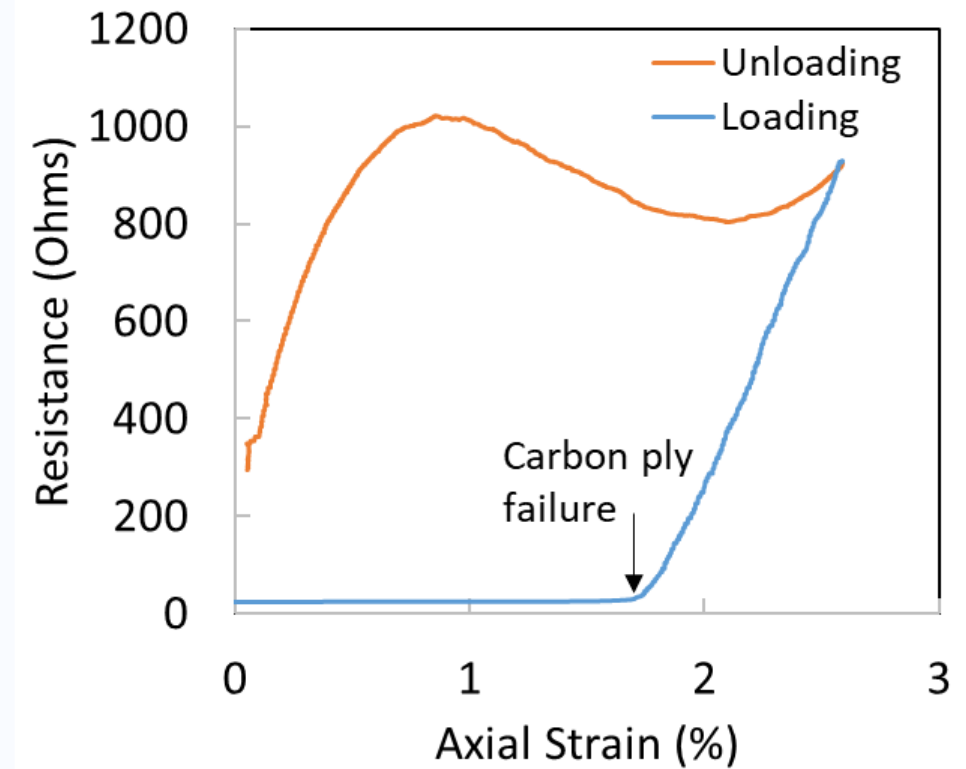
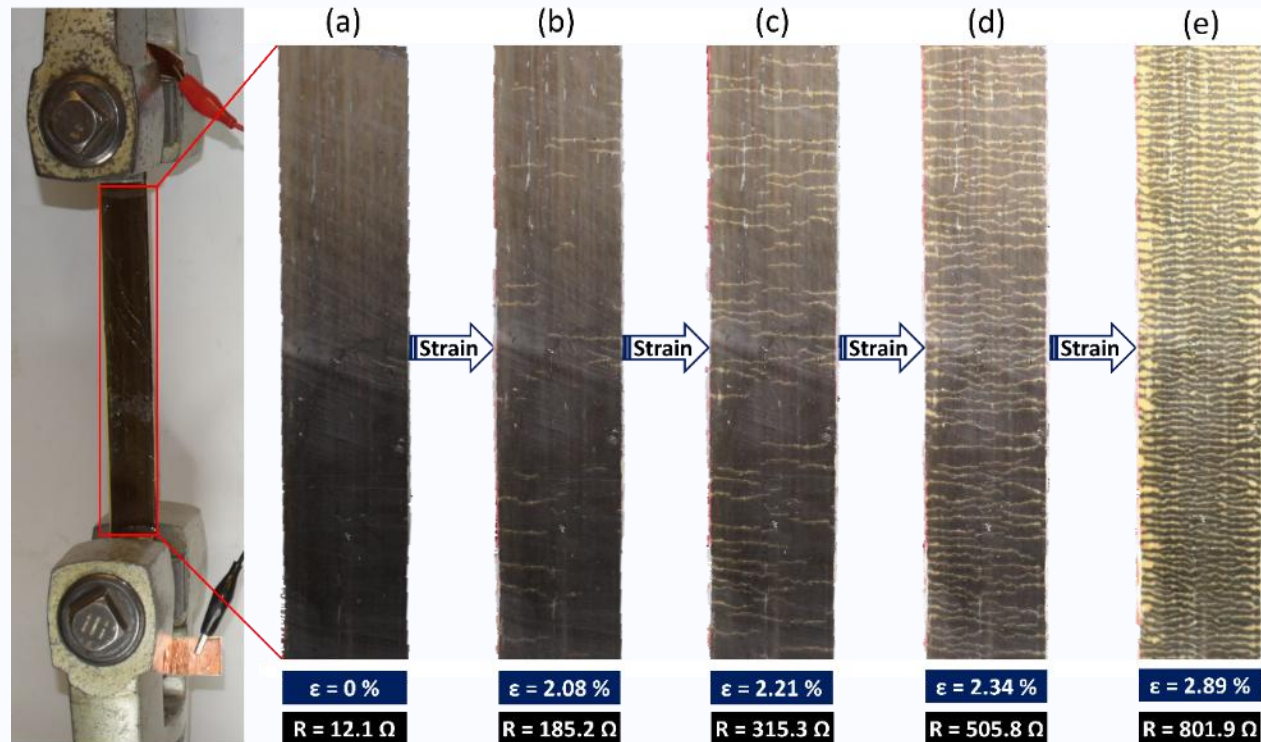
Non-piezoresistive when not overstrained



Piezoresistive when overstrained



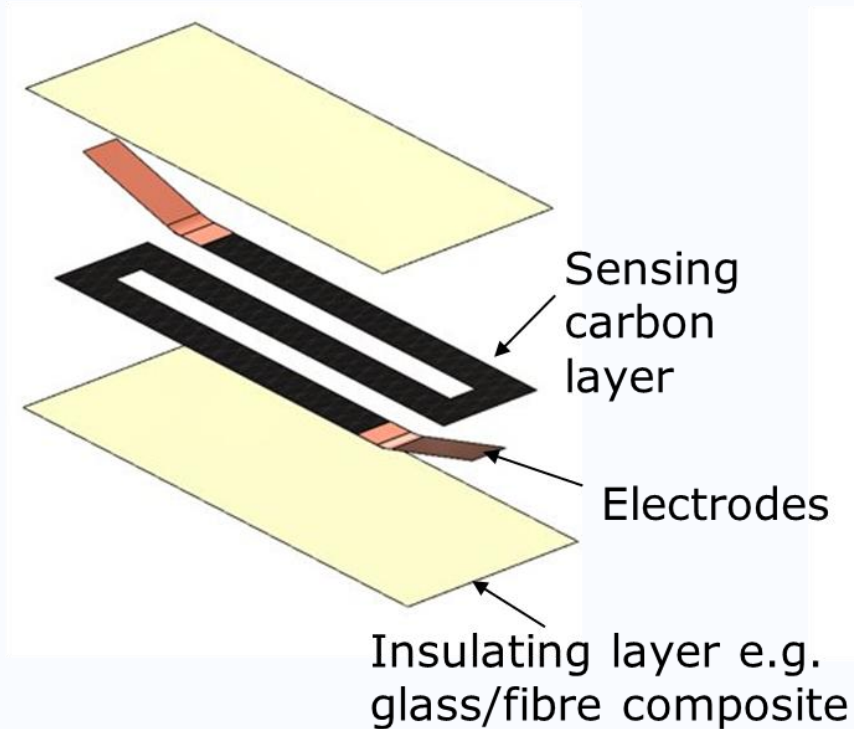
Tensile Loading-unloading



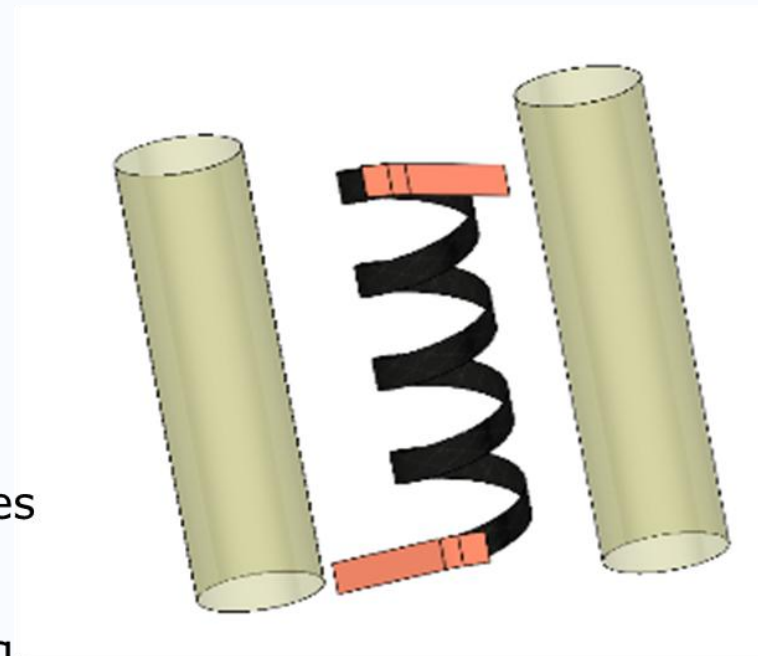
Application

- New sensor designs for identifying overstrains e.g. flat panels or cylindrical shapes

Serpentine configuration

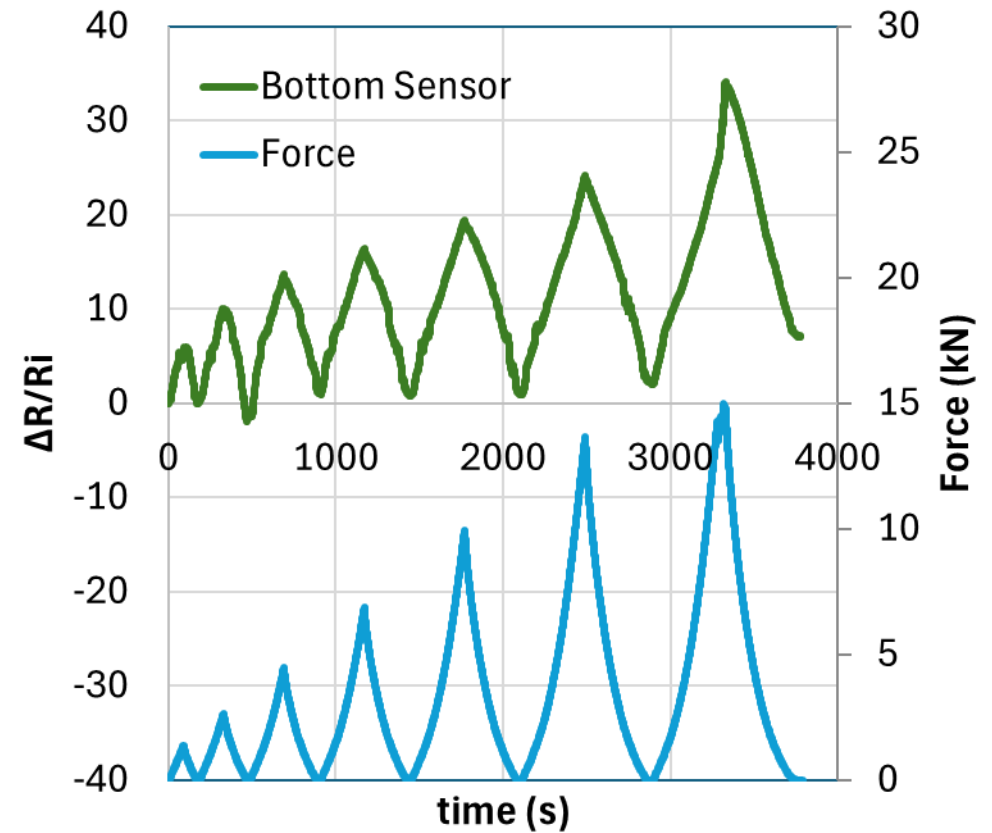
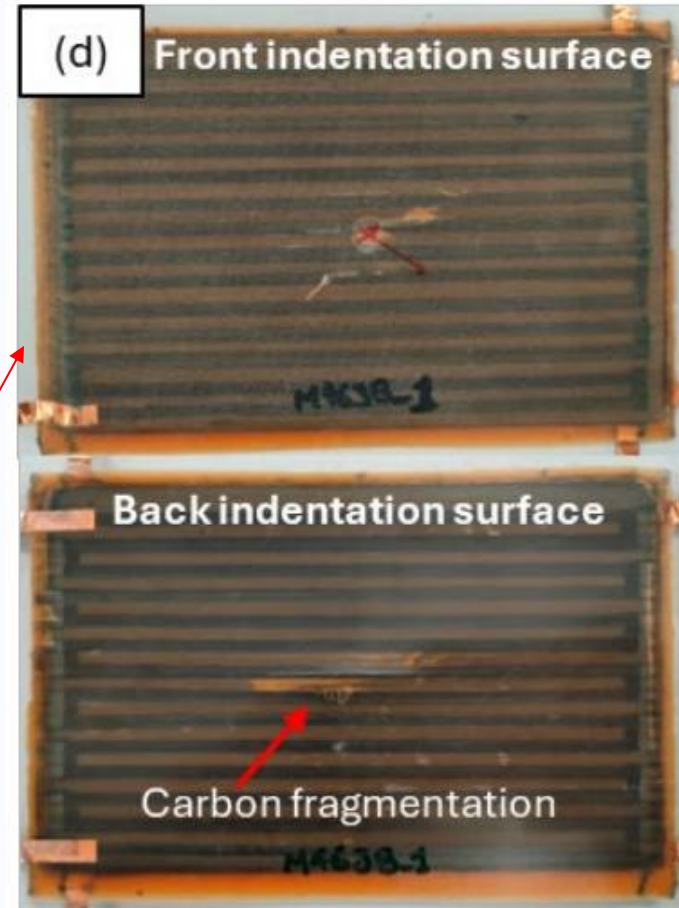
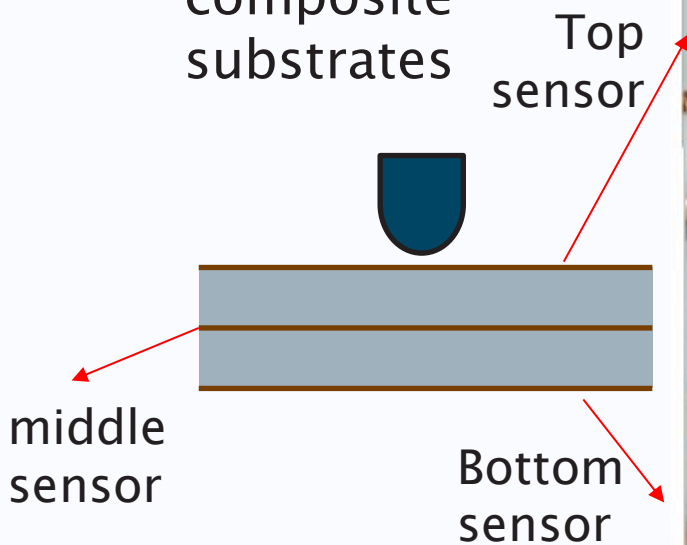


Helical configuration



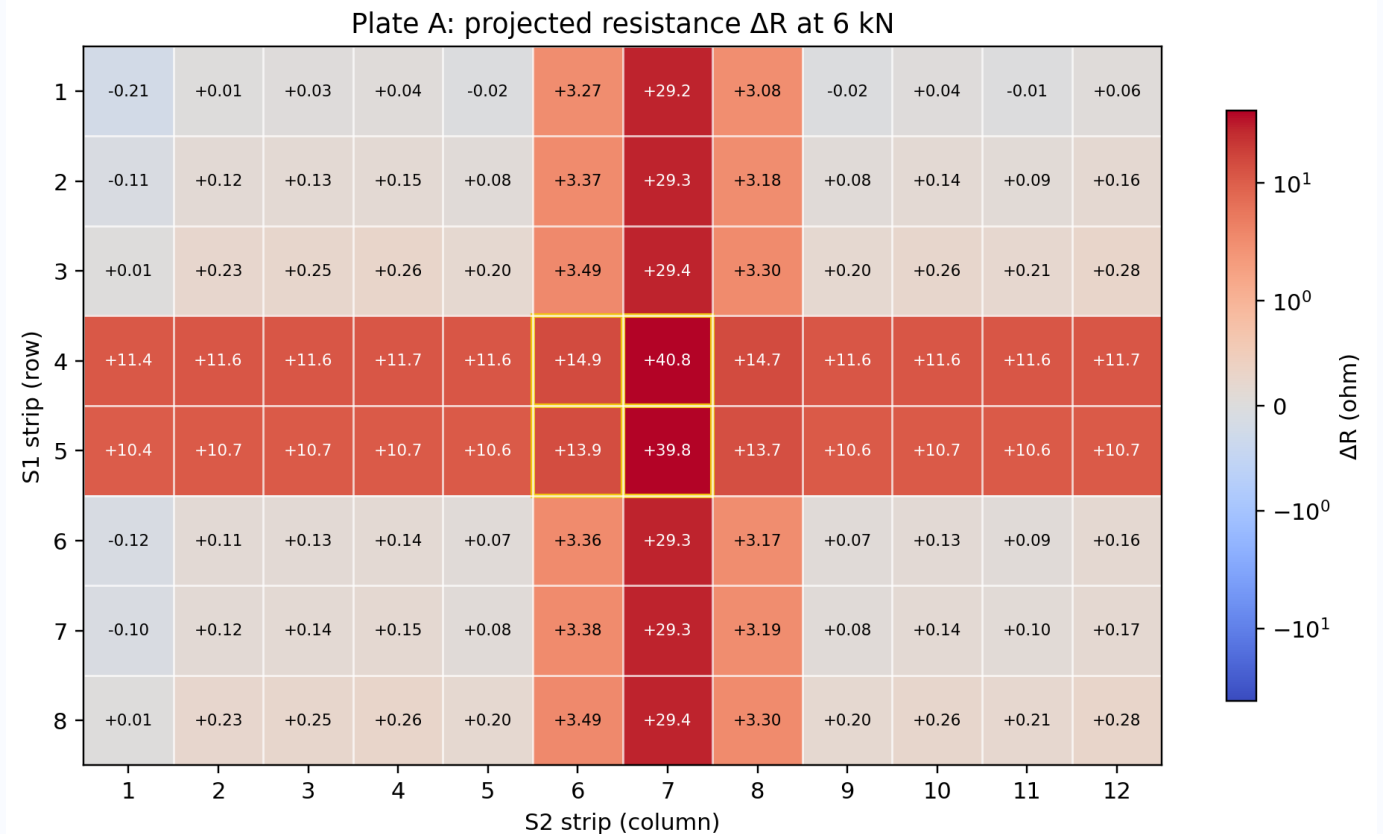
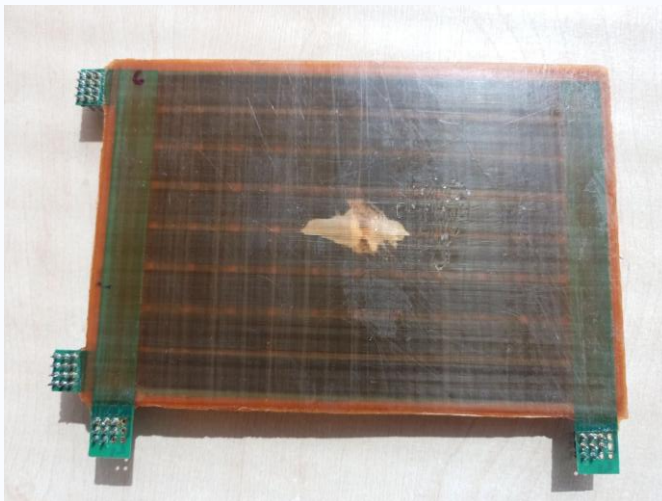
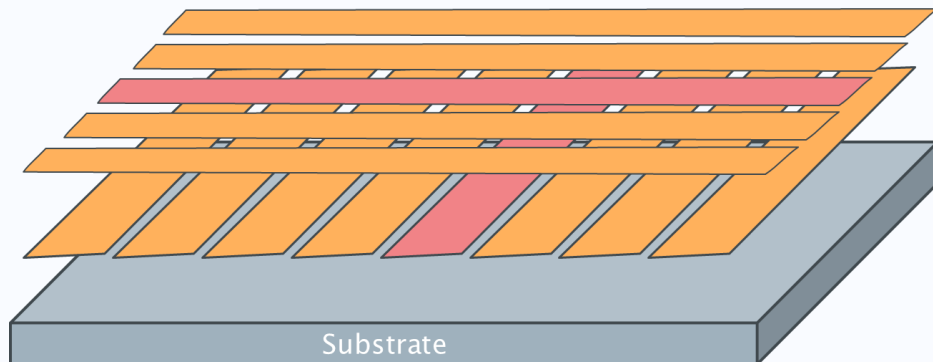
Detection of indentation overload

- Three sensors placed on the top, bottom and at the midplane of composite substrates



Full area coverage and localising the overstrain area

- Crossing Smart Tapes



Lightweight and low-cost electronics

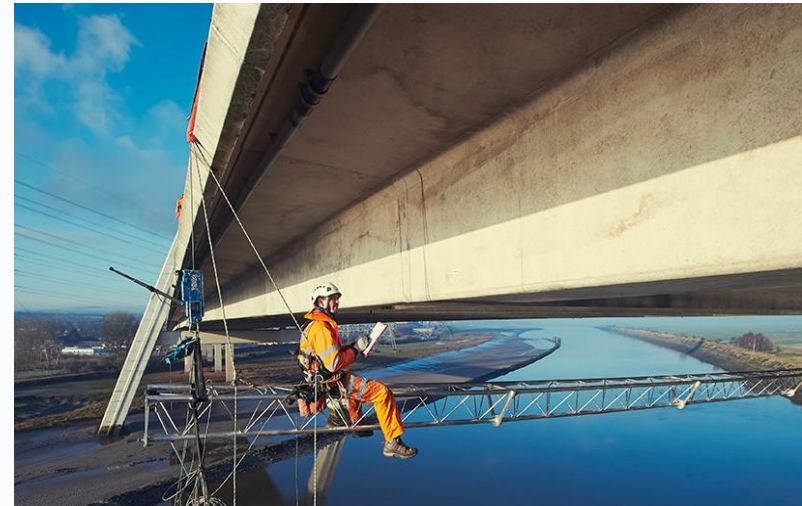
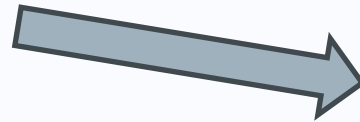
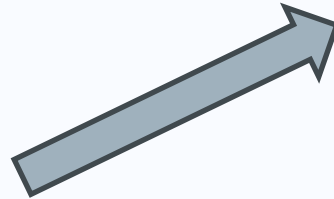
- A prototype circuit has been developed and tested electronically
- Commercially available components at low cost
- Variety of sensors included: temperature, strain gauge
- More advanced solutions feasible



Smart Tape as a product

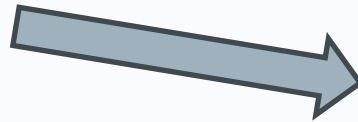
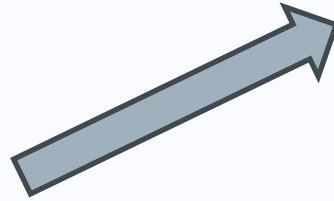
Reducing human inspection cost and downtime

- Embedding when a structure is manufactured
- Monitoring transportation, long time storage, operation



Local sensing on a patch

- Smart repair patches

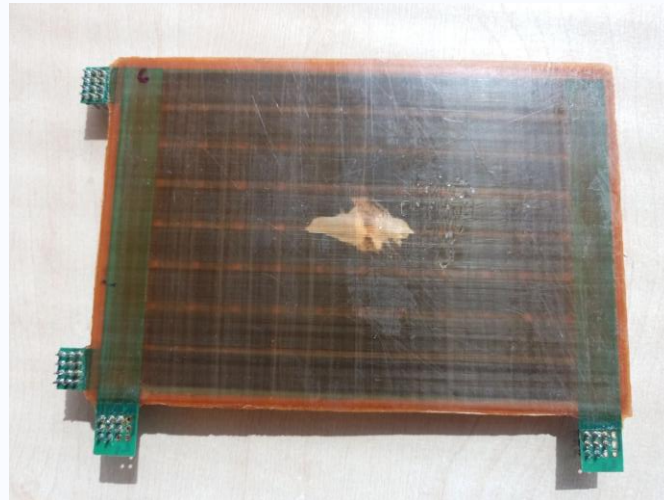
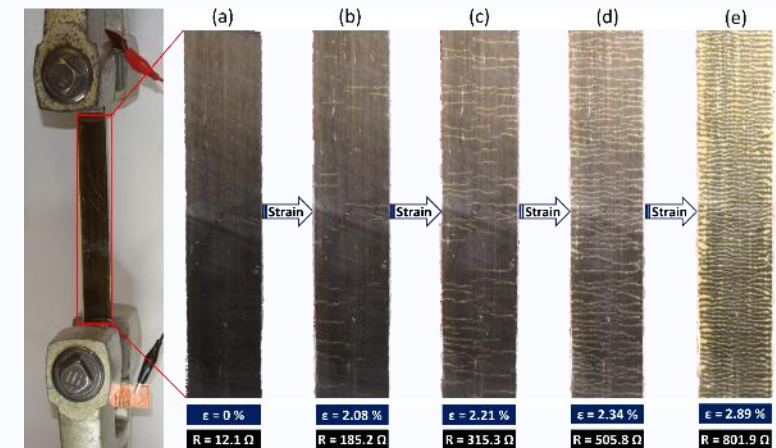


Compared to Fiber Bragg Gratings (FBG) optical fibres

	Smart Tape Overstrain sensor	FBG optical fibres
Measured parameter	overstrain	strain
Coverage	area	point
Sensor cost	Pennies / m	Pounds / strain gauge
Electronics cost	£10s	~£20k
Weight of electronics	gr	kg
Manufacturing	Compatible with filament winding	Challenging
Memory?	Yes	No
Structurally taking load	Yes	No

Overview

- Hybrid overstrain sensors provide a simple visual indication plus quantitative piezoresistive signal when structures experience overstrain events.
- Their performance has been validated on multiple configurations and loading scenarios.



YOUR QUESTIONS

Meisam Jalalvand
m.jalalvand@soton.ac.uk